

Epistemic Legitimacy as a Governance Layer for Large Language Models

Persistent Existence, Admissibility, and Lawful Continuation

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Abstract

Large language models are increasingly used in settings where an answer can become advice, a record entry, a recommendation, or a decision trigger. In such settings, the central problem is not fluency alone, but whether a model-generated conclusion is permitted by the available evidence, persistent state, and role-specific authority. This paper introduces epistemic legitimacy as a governance requirement for high-stakes LLM deployment. It presents an architecture organized around four linked claims: interpretation must be conditioned by persistent existence rather than reconstructed from local sequence alone; admissibility must be determined before output becomes consequence; refusal, suspended judgment, and clarification are valid governed outcomes; and post-refusal continuation must itself be lawfully constrained. The paper distinguishes Aurora as the broader reasoning substrate, the Persistent Existence Framework (PEF) as the persistence substrate on which meaning depends, Lens as the runtime admissibility gate, and Governor as the continuation layer that constrains what may happen after refusal, revision, or ambiguity. It also describes a provider-agnostic transport-layer implementation and a tamper-evident forensic audit record generated at decision time. This is an architecture-and-implementation paper supported by proof-of-concept demonstrations, not a benchmark paper. The claim is not that current models never produce useful answers, but that conventional stacks lack a principled deterministic mechanism for deciding whether a proposed answer is allowed to pass into consequence.

1. Introduction

Large language models now draft reports, summarize records, answer questions, and support decision-making across medicine, law, finance, customer service, and public administration. Scaling and instruction tuning have made them strikingly fluent and broadly useful [1,2,3]. Yet the most dangerous failures in high-stakes settings are not exhausted by the familiar categories of toxicity, bias, or factual hallucination. A system can produce an answer that is articulate, locally plausible, and even partially evidence-grounded while still being illegitimate in a stronger sense: it reaches a conclusion that the available structure does not permit.

This paper argues that the missing requirement is epistemic legitimacy. The key deployment question is not only whether a model can say something sensible, but whether it is allowed to say that particular thing under the relevant evidence, state, and authority conditions. Existing approaches often optimize generation first and then add prompt rules, filters, retrieval, or human review around the outside. Those

approaches can be useful, but they do not by themselves supply a deterministic governance boundary that adjudicates admissibility before model-originated output becomes consequence.

The architecture presented here begins from a different premise. Interpretation is state-dependent because entities, relations, unresolved ambiguities, and prior commitments must persist across turns if later claims are to be evaluated coherently. From that premise follow four architectural requirements: persistent existence as interpretive substrate, admissibility before consequence, refusal as a governed terminal outcome, and lawful continuation after non-admit outcomes. The implementation surface described in this paper is Aurora-Lens, a provider-agnostic governance proxy operating at the transport boundary between an application and an upstream model provider.

2. Paper Type and Scope

This is an architecture-and-implementation paper with proof-of-concept demonstrations. It is not a large-scale benchmark study, and it does not claim comprehensive empirical superiority over all existing systems across all tasks. Its aim is narrower and more structural: to define the missing governance requirement, specify the architectural components that satisfy it, and describe a working implementation surface that makes those claims operational.

The argument is supported by two kinds of material. First, it rests on a formal architectural distinction between sequence reconstruction and persistent existence. Second, it draws on proof-of-concept runtime behavior showing refusal, containment for clarification, ambiguity preservation, and audit recording in high-stakes domains. The present paper therefore should be read as a systems architecture paper that introduces a governance layer and its governing ontology, situates that layer relative to adjacent methods, and states clearly what has and has not yet been claimed.

3. The Structural Failure: Unauthorized Consequence

Public discussion often frames LLM risk in terms of whether a generated statement is factually accurate, aligned, safe, or well-behaved. Those are important concerns, but they do not fully capture the failure class at issue here. The deeper problem is unauthorized consequence: a model-generated output crosses from interpretation into action, recommendation, record formation, or decision support without a principled check on whether that commitment is admissible.

In medicine, this can appear as premature narrowing of a differential, unwarranted reassurance, or advice that exceeds evidentiary support. In legal work, it can appear as overconfident characterization of facts, authority, or procedural options under incomplete context. In finance, it can appear as unsupported projections, personalized advice, or numeric overreach. Across these settings, the failure is not simply that the system was wrong. The failure is that it was allowed to commit under conditions that should have blocked or constrained commitment.

Conventional stacks often look like this:

input -> generation -> optional filtering, retrieval, or review -> output

The architecture proposed here inserts a missing governing boundary:

input -> interpretation against persistent state -> admissibility decision -> lawful continuation

That additional step changes the status of refusal. Refusal is no longer an awkward special case. It becomes one valid outcome of correct governance when admissibility conditions are not satisfied.

4. Epistemic Legitimacy

Epistemic legitimacy is the property of a system that can determine, relative to the available evidence structure, persistent state, and authority conditions, whether a proposed conclusion is permitted, blocked, or requires clarification. It is distinct from fluency, benchmark accuracy, chain-of-thought coherence, long-context recall, or user preference optimization. A system may perform well on those dimensions and still lack a principled mechanism for deciding whether a conclusion is allowed.

A system is epistemically legitimate only if it can do at least the following:

- (1) condition interpretation on persistent state rather than local sequence alone;
- (2) preserve multiple admissible interpretations until constraints eliminate alternatives;
- (3) prohibit commitment when the evidence structure does not force resolution;
- (4) treat refusal, suspended judgment, and clarification as valid governed outcomes;
- (5) apply authority-sensitive continuation rules after non-admit outcomes; and
- (6) produce an inspectable decision state suitable for replay and audit.

The contrast can be stated simply. Intelligence asks whether a system can generate an answer. Legitimacy asks whether the system is permitted to commit to that answer under the actual governing conditions.

5. Persistent Existence and Present-Bound Interpretation

The architecture begins from a premise that most deployed LLM systems do not natively maintain: interpretation is state-dependent because entities and relations continue to exist across turns. This persistent layer is the Persistent Existence Framework, or PEF. PEF is not generic memory, not chat-history replay, and not retrieval over dead text. It is the persistence substrate in which entities, relations, ambiguities, and prior commitments remain available as present-state structure against which new text is interpreted.

This matters because conventional LLM processing tends to reconstruct a world from sequence. Meaning is inferred from whatever local context happens to be present in the prompt window. When the governing state is not explicit and persistent, continuity degrades, ambiguities are flattened, and later claims are evaluated against narrative momentum rather than against an already-existing world. PEF changes the interface. Text no longer arrives as an isolated prompt to be interpreted from scratch. It proposes mutations to an existing world.

The ontology underlying this architecture is present-bound. The system operates from the premise that there is only NOW, with temporal handling expressed through present-state representation and span transitions rather than by treating the past as a recoverable active workspace. This framing is not introduced as a general theory of consciousness. In this architecture it functions as a systems constraint: if interpretation is activation-bound, then persistence must be represented explicitly rather than assumed [4].

6. Parallel Epistemic Field and Constraint-Forced Collapse

Within the persistence substrate, the system still requires a disciplined way to handle underdetermination. That role is served by the Parallel Epistemic Field. The field maintains admissible interpretive possibilities without forcing premature collapse. Unlike chain-of-thought prompting, which externalizes one reasoning path at a time [6], the field preserves multiple admissible paths until constraints eliminate all but one.

Several properties are essential. First, non-temporal persistence: an unresolved field does not decay merely because time passes or the session continues. Second, constraint-forced collapse only: commitment is prohibited unless the available constraints eliminate alternatives. Third, admissibility gating before commitment: before any composite interpretive state is instantiated, the proposed commitment must pass an admissibility boundary. Fourth, ambiguity as a first-class state: ambiguity is not treated as an inconvenience to be smoothed away by heuristic preference. Fifth, deterministic governance: given the same relevant state and constraints, the governance layer must produce the same outcome.

These properties are not decorative. They directly address a recurrent transformer failure mode: the collapse of genuine ambiguity into a single fluent answer. In focused proof-of-concept ambiguity demonstrations used during development, transformer systems across multiple platforms repeatedly defaulted to one interpretation even when context supported more than one valid reading. The point of that demonstration is not that pronouns are the central industrial problem. The point is that premature collapse in a small linguistic case reveals the same structural tendency that becomes dangerous in clinical, legal, or financial interpretation.

7. Architectural Layers: Aurora, Lens, and Governor

The broader architecture has three distinct layers, and keeping them distinct matters both conceptually and practically.

Aurora is the broader reasoning substrate. It provides the persistence framework, governing primitives, structured state representation, and the general architecture in which admissibility can be defined. Lens is the runtime admissibility gate. It determines whether model-originated output may become consequence. Governor is the lawful continuation layer. Once Lens has admitted, contained, revised, or refused an output, Governor determines what forms of continuation remain allowed in that domain and for that authority class.

This separation matters because many adjacent systems are really filters, wrappers, dashboards, observability layers, or retrieval stacks. Those systems can be useful, but they are not equivalent to an admissibility architecture. Governance is not merely the removal of disallowed topics. It is the adjudication of whether a concrete proposed output is admissible relative to persistent state, authority, and consequence.

A compact governing expression for continuation is:

domain × authority class → allowed continuation set

That expression captures a crucial design point. A general-purpose model that has been refused on a clinical question is not thereby licensed to continue by softening its language. The next permitted move must itself be determined by domain and authority constraints.

8. Runtime Operation of Aurora-Lens

Aurora-Lens is the deployable runtime expression of the admissibility layer. It operates as a provider-agnostic governance proxy between an application and an upstream model provider. The application continues to interact through a standard chat-completions style interface; governance is applied at the transport boundary rather than by retraining or replacing the underlying model.

At runtime, the system performs the following sequence. It intercepts the request before it reaches the model provider. It interprets the request against the relevant persistent state. It evaluates admissibility, determining whether the requested or proposed output may pass, must be revised, must be contained for clarification, or must be refused. When admissible to proceed, it forwards the request under governance conditions. It then verifies model-originated claims against persistent state, authority, and domain constraints. Finally, it records the governing decision state in a forensic audit record.

The adapter layer is deliberately minimal. Governance does not depend on provider-specific benevolence or model-internal self-policing. The implementation surface described here has been used to demonstrate cross-domain governance patterns including refusal, containment for clarification, ambiguity preservation, and decision-state logging. Operational alignment with due-diligence style governance requirements is discussed separately in related public work [5].

9. Lawful Continuation After Non-Admit Outcomes

A system that can refuse but has no constrained continuation strategy is only half-built. Real deployment requires a second boundary after refusal, revision, or ambiguity. That is the role of Governor.

Lawful continuation may include, depending on domain and authority, clarification requests, structured escalation, explanation of refusal, handoff to a qualified human, or bounded reformulation that does not widen the denied commitment. What it may not include is quiet re-entry into the same forbidden commitment through softer wording, partial hedging, or conversational drift.

This layer is essential because otherwise refusal becomes performative. The system says no in one line and then erodes its own boundary in the next. Governor preserves the consequence boundary established by Lens instead of treating refusal as a cosmetic interruption.

10. Forensic Auditability at Decision Time

Auditability is meaningful only if it captures the state that actually produced the decision. Post-hoc dashboards and after-the-fact summaries are useful for observability, but they do not establish what the system was allowed to do at the moment consequence became possible. Aurora-Lens therefore writes a tamper-evident forensic audit record at decision time.

The record is designed to answer not merely what the model said, but whether it was permitted to say it, under what governing conditions, and with what consequence status. This matters in regulated settings for at least three reasons. First, organisations need to reconstruct why a system acted or refused. Second, auditors and investigators need replayable evidence rather than narrative reconstruction. Third, the existence of a replayable consequence boundary disciplines system design: once every consequential decision must be reproducible, sloppy heuristic collapse becomes visible.

Several existing approaches address parts of the governance problem, but none by themselves supply epistemic legitimacy in the sense defined here.

Prompt engineering and chain-of-thought prompting can improve reasoning quality or encourage caution, but they do not by themselves enforce admissibility [6]. Retrieval-augmented generation can improve grounding to documents, but a conclusion can still exceed what the retrieved material justifies [7]. Instruction tuning and preference optimization can improve helpfulness and reduce some harmful behaviors, but they do not create a separate deterministic governance path that adjudicates commitment before consequence [3,8]. Tool-use architectures can extend capability, but deciding when an output may pass remains a governance problem even when models can call external systems [9].

The point is not that these methods are useless. The point is that they operate at different layers. This architecture governs the AI answer, not merely the model prompt, the retrieved context, or the surrounding observability stack.

12. Contributions

The principal contributions claimed in this paper are these.

First, it defines epistemic legitimacy as a governance requirement for LLM deployment in high-stakes settings.

Second, it specifies persistent existence, rather than local sequence reconstruction, as the interpretive substrate required for coherent admissibility judgments across turns.

Third, it describes the Parallel Epistemic Field as a mechanism for preserving admissible alternatives until constraint-forced collapse is justified.

Fourth, it distinguishes Aurora, Lens, and Governor as separate architectural layers corresponding respectively to reasoning substrate, admissibility decision, and lawful continuation.

Fifth, it treats refusal, suspended judgment, and clarification as first-class governed outcomes rather than failures of user experience.

Sixth, it places forensic auditability at decision time rather than in retrospective reporting alone.

13. Limitations and Next Steps

Several limits should be stated plainly. This paper does not yet provide a large benchmark suite, a formal complexity analysis of all primitives, or a broad comparative evaluation across deployment stacks. It also does not argue that all existing safety or alignment methods are obsolete. Many will remain useful components within larger systems.

The claim here is narrower and more exacting: conventional LLM deployments lack a principled deterministic mechanism for deciding whether a proposed answer is allowed to pass as consequence under persistent state, evidentiary constraint, and authority conditions. The architecture described here is intended to supply that missing layer.

Immediate next steps include broader empirical evaluation, formalization of the continuation matrix across domains and authority classes, and publication of additional proof-of-concept demonstrations showing how decision-time audit and admissibility interact in domain-specific workflows.

14. Conclusion

The central problem in high-stakes LLM deployment is not only that models may be wrong. It is that they are routinely allowed to commit under conditions where commitment is not structurally justified. Epistemic legitimacy names that missing requirement.

Persistent existence conditions interpretation. The Parallel Epistemic Field preserves admissible alternatives until collapse is forced. Lens decides whether model-originated output may become consequence. Governor constrains what may lawfully happen after refusal, revision, or ambiguity. Decision-time forensic audit preserves the state that produced the decision.

Together these components define a governance architecture rather than a prompting style. The result is not a more polite model. It is a system that can refuse illegitimate commitment and preserve consequence boundaries in settings where fluent overreach is unacceptable.

References

- [1] Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33.
- [2] Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*.
- [3] Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training Language Models to Follow Instructions with Human Feedback. *Advances in Neural Information Processing Systems*, 35.
- [4] Stokes, M. (2026). Present-Centered Cognition Model (PCCM): A Cross-Substrate Framework for Understanding Consciousness, Presence, and Latent States in Humans and Machines. *Zenodo*. <https://doi.org/10.5281/zenodo.18976303>
- [5] Stokes, M. (2026). Operational Alignment of Aurora-Lens with OECD Due Diligence Guidance for Responsible AI. *Zenodo*. <https://doi.org/10.5281/zenodo.18719033>
- [6] Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *Advances in Neural Information Processing Systems*, 35.
- [7] Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33.
- [8] Bai, Y., Kadavath, S., Kundu, S., et al. (2022). Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*.
- [9] Schick, T., Dwivedi-Yu, J., Dessi, R., et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. *Advances in Neural Information Processing Systems*, 36.